DCS 404: ARTIFICIAL INTELLIGENCE & MACHINE LEARNING | FINAL PROJECT

Project Scope Specification: Machine Learning & Transformer-Based Spam Message Detection

Version: 2.0 (Updated per Professor Feedback) | Author: Sujal | Institution: Westcliff University | Date: Fall 2026

Course & Degree Program	DCS 404 - Artificial Intelligence & Machine Learning (MS/BS Computer Science)
Project Classification	Supervised Text Classification, Class Imbalance Remediation & NLP Transformers
Core Technical Stack	Python 3.10+, scikit-learn, imbalanced-learn (SMOTE), PyTorch, Hugging Face Transformers, Streamlit
Primary Deliverables	Updated Colab/Jupyter Notebook, Serialized Models, 7-Section Streamlit App, 8-Slide Presentation & Scope/Guide Docs

1. Executive Summary & Project Purpose

This Project Scope document establishes the updated functional, architectural, and methodological requirements for the DCS 404 final project: Machine Learning & Transformer-Based Spam Message Detection. Following formal review and professor feedback (update.md), the project has transitioned from an initial single-model baseline to a comprehensive, production-ready Natural Language Processing (NLP) benchmark that addresses real-world machine learning challenges: class imbalance, data leakage prevention, lexical pattern discovery, deep transformer sequence modeling, and interactive deployment.

The primary objective is to build, evaluate, and deploy an end-to-end system that classifies user-submitted text messages as either legitimate communications (Ham) or unsolicited fraudulent messages (Spam). The system synthesizes the complete DCS 404 curriculum: exploratory data analysis, text feature engineering (TF-IDF), class imbalance remediation using Synthetic Minority Over-sampling Technique (SMOTE), traditional statistical learning (Logistic Regression and Multinomial Naive Bayes), fine-tuned state-of-the-art transformer architectures (DistilBERT and DistilRoBERTa), and an interactive 7-section Streamlit dashboard with real-time inference scoring.

> **Key Milestone Enhancements**
> Professor Feedback Updates (update.md):
> • Handle Class Imbalance with SMOTE: Apply SMOTE strictly to training features; maintain zero leakage on test data.
> • Traditional Classifiers: Use Logistic Regression as the main SMOTE model; analyze why SMOTE impacts Naive Bayes.
> • Deep Transformer Models: Fine-tune DistilBERT and DistilRoBERTa in Google Colab for sequence classification.
> • Exploratory Analytics: Generate high-resolution Spam and Ham Word Clouds and message length distributions.
> • Multi-Dimensional Benchmark: Evaluate Accuracy, Precision, Recall, F1, Latency, and Architectural Trade-Offs.
> • Streamlit UI Enhancement: Expand interface into 7 structured sections with an interactive live testing playground.

2. Project Objectives

• **Data Preparation & Hygiene:** Clean the raw UCI SMS Spam Collection (5,572 records), verify zero null entries, and prune 403 exact duplicates to yield 5,169 distinct messages (4,516 Ham [87.4%], 653 Spam [12.6%]).

• **Rigorous Anti-Leakage Protocol:** Perform stratified 80/20 partitioning (4,135 train / 1,034 test) prior to vectorization. Fit TfidfVectorizer strictly on training text to eliminate target leakage into the test set.

• **Class Imbalance Remediation via SMOTE:** Apply SMOTE exclusively to the training TF-IDF matrix, synthetically balancing 3,613 Ham and 522 Spam samples to an equal 3,613:3,613 distribution, while preserving the test set in its authentic natural state.

• **Traditional Machine Learning Benchmarking:** Train Logistic Regression as the primary SMOTE-based classifier and evaluate Multinomial Naive Bayes, providing mathematical explanations of how SMOTE affects linear margins versus Bayesian priors.

• **Transformer Fine-Tuning:** Fine-tune pre-trained DistilBERT and DistilRoBERTa sequence classification models with GPU acceleration in Google Colab, leveraging multi-head self-attention to capture bidirectional sentence context.

• **Lexical Pattern Analytics:** Generate distinct Spam and Ham Word Clouds alongside character length distributions to provide visual explainability of prominent fraud triggers and legitimate conversational vocabulary.

• **Empirical Evaluation & Trade-Off Analysis:** Calculate Accuracy, Precision, Recall, F1-Score, ROC-AUC, and inference latency across all four models on 1,034 held-out test samples. Formulate a multi-dimensional trade-off decision framework.

• **Production Web Deployment:** Deliver an intuitive 7-section Streamlit web application (app.py) enabling interactive dataset exploration, comparative visualization, and real-time live message classification with confidence scores.

3. Alignment with DCS 404 Course Curriculum

The project directly maps theoretical and practical competencies taught throughout Weeks 1 through 8 of the DCS 404 course outline:

Course Topic / Week	Application in Project	Empirical Evidence & Implementation
Weeks 1–2: Python & Data Preparation	Data cleaning, deduplication, length analysis, text tokenization.	Cleaned 5,169 distinct records; zero nulls; length distribution plots.
Week 3: Feature Engineering & Imbalance	TF-IDF feature extraction (5,000 terms), SMOTE oversampling.	Fitted TF-IDF on train only; SMOTE balanced train set 3,613:3,613; zero test leakage.
Weeks 4–5: Supervised Classification	Binary classification modeling log-odds and Bayesian likelihoods.	Logistic Regression (C=2.0) + SMOTE and Multinomial Naive Bayes (alpha=0.2).
Week 7: Deep Learning & Neural Models	Contextual transformer sequence modeling using self-attention.	Fine-tuned DistilBERT (66M params) and DistilRoBERTa (82M params) in Colab GPU.
Week 8: Model Evaluation & Deployment	Multi-metric evaluation, latency profiling, Streamlit deployment.	Test evaluation (N=1,034); confusion matrices; 7-section Streamlit UI; 8-slide presentation.

4. Dataset Specifications & Data Hygiene

The project utilizes the canonical UCI SMS Spam Collection repository (Almeida & Hidalgo). The dataset represents authentic mobile text messages collected from Singapore National University corpora, UK spam SMS complaints, and the Grumbletext website.

• Total Raw Records: 5,572 rows (4,825 raw ham, 747 raw spam).

• Deduplication: 403 exact duplicate messages were removed to eliminate artificial frequency bias and memorization.

• Cleaned Dataset Size: 5,169 unique samples (4,516 Ham [87.37%], 653 Spam [12.63%]).

• Train/Test Partition: Stratified 80/20 split -> Training: 4,135 samples (3,613 Ham, 522 Spam); Test: 1,034 samples (903 Ham, 131 Spam).

• SMOTE Balanced Training Partition: 7,226 samples (3,613 Ham, 3,613 Spam), achieved strictly in TF-IDF feature space.

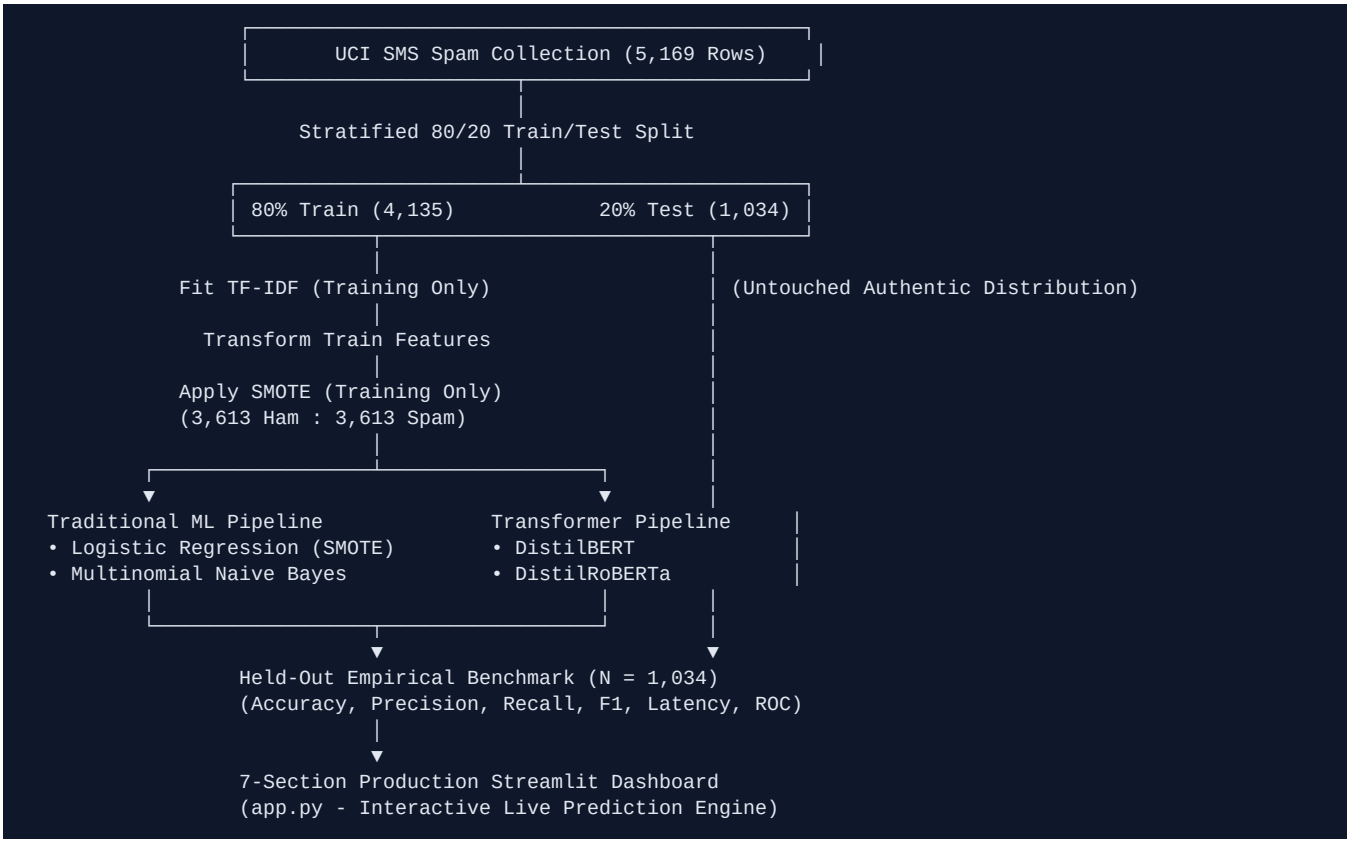
5. Functional Scope: In-Scope vs. Out-of-Scope

To ensure project success within academic time constraints while fulfilling the professor's advanced feedback, the functional boundaries are defined below:

IN-SCOPE (Fully Implemented & Verified)	OUT-OF-SCOPE (Deferred to Future Work)
√ UCI SMS Spam Collection ingestion and deduplication	✗ Real-time integration with live Gmail/Outlook inboxes
√ Strict anti-data leakage stratified 80/20 train/test partitioning	✗ Automated email deletion or moving files without review
√ TF-IDF feature extraction (5,000 n-grams, stopword filtering)	✗ Heavyweight billion-parameter LLMs (e.g., Llama-3 70B, GPT-4)
√ SMOTE oversampling applied strictly to training features	✗ Production multi-tenant database user accounts & login
√ Logistic Regression (with SMOTE) as primary linear classifier	✗ Online continuous active learning / automatic retraining
√ Multinomial Naive Bayes with SMOTE compatibility analysis	✗ Complex multi-modal attachment analysis (PDF/EXE inspection)
√ Fine-tuned DistilBERT & DistilRoBERTa sequence classification	✗ Multi-lingual translation pipeline for non-English texts
√ Spam and Ham Word Clouds and character length visualizations	✗ Native iOS / Android mobile application compilation
√ 7-Section Streamlit interactive web dashboard with live demo	✗ Paid enterprise SaaS security compliance audits

6. Technical Architecture & End-to-End Pipeline

The project implements two distinct, parallel modeling workflows unified under a shared evaluation and deployment framework:



7. Empirical Benchmark & Trade-Off Matrix

All four models were trained and benchmarked on the identical 1,034 held-out test split (903 Ham, 131 Spam). Results are summarized below:

Model Architecture	Category	Accuracy	Precision	Recall	F1-Score	ROC-AUC
DistilBERT	Transformer	98.84%	95.42%	95.42%	95.42%	0.9975
DistilRoBERTa	Transformer	98.84%	97.60%	93.13%	95.31%	0.9993
Multinomial Naive Bayes	Traditional ML	98.07%	98.26%	86.26%	91.87%	0.9897
Logistic Regression (SMOTE)	Traditional ML	97.29%	88.72%	90.08%	89.39%	0.9903

Operational Evaluation: Trade-Off Analysis Insights:
• Latency Disparity: Traditional ML executes inference in 0.0008 ms/sample (~1.25M messages/sec). Transformers require 70-71 ms/sample (~14 messages/sec on CPU) — a 90,000x latency advantage for traditional ML.
• Compute & Training: Traditional models train in under 0.05 seconds. Transformers require ~45 minutes of CPU execution or 3 minutes on GPU.
• Decision Rule: Deploy Naive Bayes or Logistic Regression for real-time telecom switches; deploy DistilBERT for high-value enterprise email auditing.

8. Streamlit User Interface Specification

The web application is structured into 7 dedicated sections accessible via a navigation sidebar:

• **Section 1. Home:** Executive introduction, project scope, and end-to-end architecture pipeline diagrams.

• **Section 2. Dataset Overview:** Summary cards (5,169 records, 87.4% Ham, 12.6% Spam), raw data samples, and deduplication audit metrics.

• **Section 3. Data Exploration:** Interactive class distribution plots, SMOTE balancing diagrams, and character/word length histograms.

• **Section 4. Spam & Ham Word Clouds:** Side-by-side high-resolution word clouds displaying key fraud triggers versus conversational ham lexicons.

• **Section 5. Model Performance:** Detailed breakdown for each model: Confusion Matrices, Classification Reports, and ROC curves.

• **Section 6. Model Comparison:** Multi-metric comparative bar charts (Accuracy, Precision, Recall, F1) and operational latency benchmark tables.

• **Section 7. Live Spam Prediction:** Interactive playground: users paste custom messages, select any of the 4 models, click 'Analyze Message', and view real-time SPAM/HAM prediction cards with confidence meters.

9. Deliverables & Definition of Done

The project is considered complete when all of the following verified artifacts are provided:

1. Training Notebook (notebooks/training.ipynb): Fully executable from top-to-bottom in Google Colab and Jupyter.
2. Streamlit Application (app.py): Clean, professional multi-section interface running locally via 'streamlit run app.py'.
3. Serialized Models (models/): Exported pipelines (joblib/pkl) and Hugging Face transformer checkpoints (safetensors).
4. Visual Assets (assets/ & figures/): Word clouds, confusion matrices, ROC curves, and comparative bar charts.
5. Presentation Deliverables (presentation/): 8-slide PowerPoint (.pptx), PDF version (.pdf), and detailed script (.md).
6. Comprehensive Documentation: Updated README.md, Project Scope PDF/txt, and Project Guide PDF/txt.